
Modular Programming – Practical Work

Week 2

Recap exercises from 631-1

Notations: W1Q1 = Week 1, Questions of type 1

Question types:

- Type 1: theory, concepts *knowledge*
- Type 2: understanding typically based on code reading *understanding*
- Type 3: producing yourself code *know-how*

A moodle deposit space is open every week for you to upload your solutions. Solutions of the exercises are provided one week later.

Important points:

- 602_F and 602_D classes should upload solutions **before Tuesday 24h00 the following week**
- **2 or 3 practical work will be corrected** and will count for one third of the intermediate test note

Conventions about fonts:

- Lucida Console is used for Java reserved commands
- Courier is used for the code
- Arial is used for the text

W2Q1 – 1

1. What are the three editions of Java, what are their respective target applications ?

The 3 editions are:

- J2ME (mobile edition for mobile equipment development)
- J2SE (standard edition for regular PC or laptop application development)
- J2EE (enterprise edition for enterprise grade type of applications including client-server architecture for example)

2. What is the difference between the JDK, the JRE and the Java API ?

The JDK includes the JRE that includes the Java API. The Java API contains all the Java packages that come as standard in the given edition. The JRE includes the Java API plus the Java Virtual Machine (to run an application) plus some deployment tools. The JDK includes the JRE plus tools for compiling (javac), documenting (javadoc), etc.

3. Explain the slogan created by Sun Microsystems for Java : « Write once, run anywhere »

Unlike C or C++, a Java program is not directly compiled into native instructions. Instead, the Java program is first compiled into an intermediary language called bytecode. The bytecode can then be executed by the so-called virtual machine, able to interpret the bytecode. Any computer where a virtual machine is implemented can run Java program.

W2Q2 – 1 Vehicle

1. Read the **Vehicle.java** code below.
 - a. Explain the difference between line 11 and 13

The access modifier is `public` for line 11, which means that the instance variable `numberOfPassengers` can be seen and modified outside of the class `Vehicle`. The access modifier is `private` for line 13, which means that the instance variable `speed` can only be modified internally in the methods of class `Vehicle` and not externally from another class.

- b. What does mean 'this' in line 16

The keyword `this` is a reference to the current instance of the class (to the current object). In the case of line 16, it is used to differentiate between the variables passed in as arguments of the method and the instance variables that have the same name.

- c. Explain the signature of `computeDistance()` in line 38 (*using the right terminology*)

```
public int computeDistance(int timeSpentInVehicle)
```

The method named `computeDistance` is a public method, taking as input parameter an integer value and returning an integer value as output.

2. Read the **VehicleTestWrong.java** code. Explain what is wrong with this code

Lines 18, 19 and 26 are wrong.

In line 18 and 19, the instance variable `speed` and `name` can not be accessed from outside the class as they are declared private.

In line 26, the method `validateSpeed` can not be accessed from outside the class as it is declared private.

```

1  /*
2   * Practical Work Week 2 - reading and understanding code
3   * - Explain the difference between line 11 and 13.
4   * - What means 'this' in line 16
5   * - Explain the signature of computeDistance in line 38
6   *
7   * Author: Jean Hennebert 2009
8   */
9  public class Vehicle {
10
11      public int numberOfPassengers;
12      private String name;
13      private int speed;
14
15      public Vehicle(String name, int speed) {
16          this.name = name;
17          this.speed = speed;
18          validateSpeed();
19      }
20
21      public String getName() {
22          return name;
23      }
24
25      public void setName(String name) {
26          this.name = name;
27      }
28
29      public int getSpeed() {
30          return speed;
31      }
32
33      public void setSpeed(int speed) {
34          this.speed = speed;
35          validateSpeed();
36      }
37
38      public int computeDistance(int timeSpentInVehicle) {
39          int distance;
40          distance = speed * timeSpentInVehicle;
41          return distance;
42      }
43
44      public boolean isRunning() {
45          if(speed!=0) {
46              return true;
47          } else {
48              return false;
49          }
50      }
51
52      private void validateSpeed() {
53          if(speed<0) {
54              System.out.println("A negative speed is not allowed, setting it to 0");
55              speed = 0;
56          }
57          if(speed>120) {
58              System.out.println("Your speed is too high, setting it to 120!");
59              speed = 120;
60          }
61      }
62
63  }

```



```
1 /*
2  * Practical Work Week 2 - reading and understanding code
3  * What is wrong with the statements in the main of class
4  * VehicleTestWrong?
5  *
6  * Author: Jean Hennebert 2009
7  */
8 public class VehicleTestWrong {
9
10     public static void main(String args[]) {
11
12         Vehicle myBike = new Vehicle("Blue bike", 20);
13         Vehicle myCar = new Vehicle("Red alfa", 100);
14
15         myBike.numberOfPassengers = 1;
16         myCar.numberOfPassengers = 4;
17
18         myBike.speed = 25;
19         myCar.name = "Yellow smart";
20
21         myCar.setName("Green Porsche");
22
23         myBike.setSpeed(-10);
24         myCar.setSpeed(220);
25
26         myCar.validateSpeed();
27
28     }
29
30 }
```

W2Q3 – 1 Circle

Modify the following code to comply with the code writing conventions of 632-1 class. Use the document [Code conventions PDF](#) available on CyberLearn. Follow strictly the guidelines highlighted in yellow.

```
1 package TP2;
2 import java.util.Scanner;
3 public class Circle {
4     private int x ;
5     final double pi = Math.PI ;
6     public void readcircle(int nX, int nY, int nR)
7     {
8         x = nX ; y = nY ; radius = nR ;
9     }
10    public Circle(int x, int y, int radius)
11    {
12        this.x = x ;
13        this.y = y ;
14        this.radius = radius ;
15    }
16    private int y ;
17    public boolean compare(Circle other){
18        if(this.x == other.x && this.y == other.y && this.radius == other.radius)
19            return true ;
20        return false ;
21    }
22    Scanner input = new Scanner(System.in) ;
23    public void lireCercle()
24    {
25        System.out.println("x : ");
26        x = input.nextInt() ;
27        System.out.println("y : ");
28        y = input.nextInt() ;
29        System.out.println("Radius : ");
30        radius = input.nextInt() ;
31    }
32    private int radius ;
33    public void move(int nX, int nY){
34        if (nX > 0)
35            x = nX ;
36        if (nY > 0)
37            y = nY ;
38    }
```

```

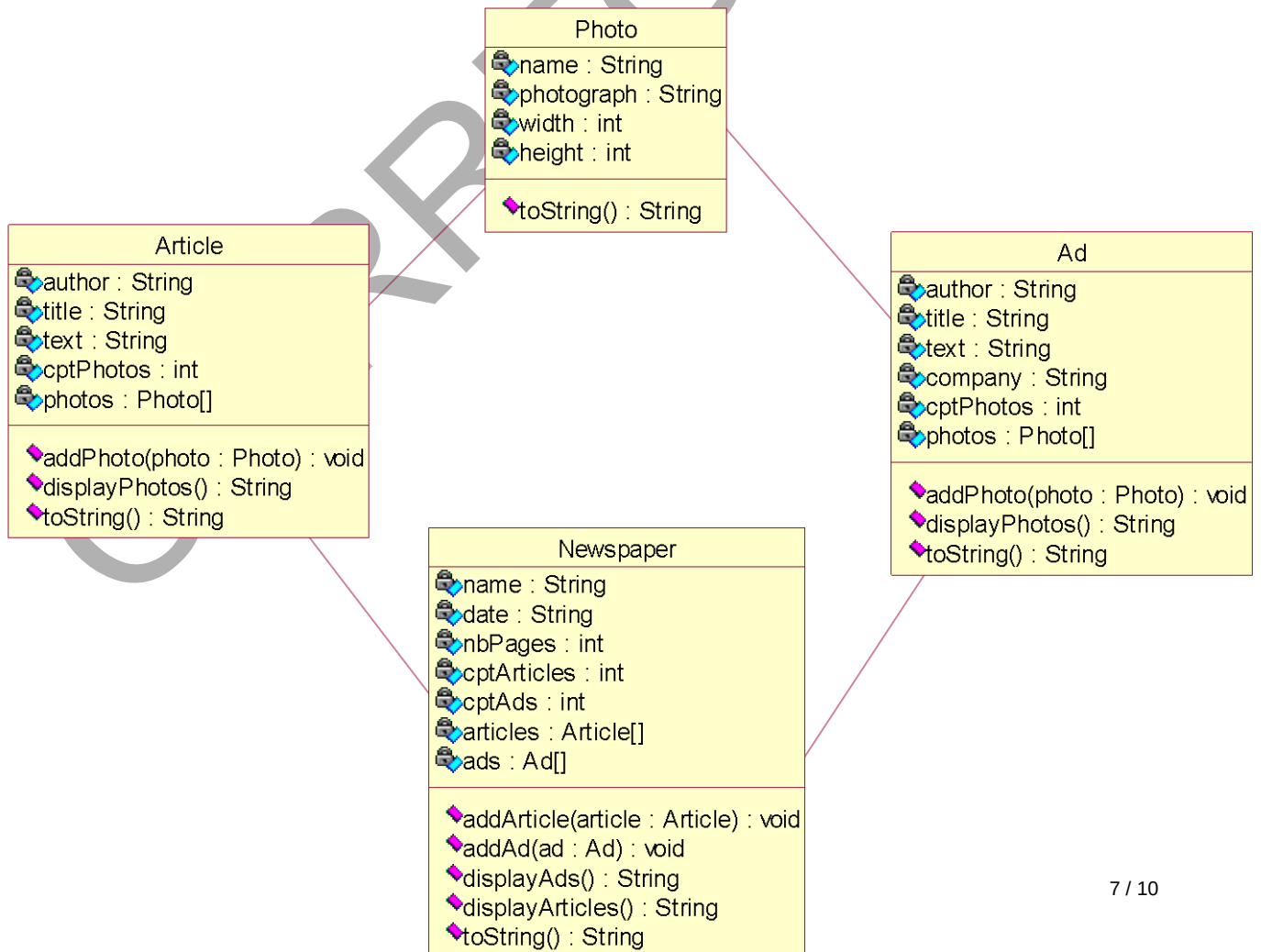
39 public void setradius(int NewRadius)
40 {
41     radius = NewRadius ;
42 }
43 public void display()
44 {
45     System.out.println("Value of x and y : (" + x + "," + y + ")" ) ;
46     System.out.println("Radius : " + radius);
47 }
48 public double perimeter()
49 {
50     return 2*pi*radius ;
51 }
52 public String toString(){
53     return "Value of x and y : (" + x + "," + y + ")\nRadius: " + radius ;
54 }
55 }

```

Circle_Correct.java

W2Q3 – 2 News paper

Create a set of classes to represent the class diagram illustrated below.



We provide you a Test class with a main method as follows. The rule is that you can't change this main method and should design your classes and methods accordingly.

```
/**
 * PW Week 2
 * Exercise W2Q3-2
 *
 * Class: Test.java
 * Description: Testing the cration of Newspapers
 * Auteur: Mélanie Zufferey and Jean Hennebert
 * Date of creation: 20.02.2009
 * Last update: 01.03.2010
 *
 */

public class TestNewspaper{

    public static void main(String[] args) {

        //Article constructor with author, title, text, number of photos
        Article article1 = new Article("Zuber", "Titre 1", "BlaBla 1", 2);
        Article article2 = new Article("Emery", "Titre 2", "BlaBla 2", 1);
        Article article3 = new Article("Zufferey", "Titre 3", "BlaBla 3", 1);
        Article article4 = new Article("Epiney", "Titre 4", "BlaBla 4", 0);
        //Ad constructor with author, title, text, company, number of photos
        Ad ad1 = new Ad("Ruedin", "Gruyeres AOC", "BlaBlaBla", "Migros", 2);
        Ad ad2 = new Ad("Zuber", "NF", "BlaBlaBla", "Nouvelliste", 2);
        Ad ad3 = new Ad("Antille", "Edelweiss", "BlaBlaBla", "ChocVillars", 1);

        Photo photo1 = new Photo("Photo 1", "Constantin", 5, 15);
        Photo photo2 = new Photo("Photo 2", "Constantin", 10, 13);
        Photo photo3 = new Photo("Photo 3", "Constantin", 15, 5);
        Photo photo4 = new Photo("Photo 4", "Constantin", 9, 15);
        Photo photo5 = new Photo("Photo 5", "Constantin", 5, 15);
        Photo photo6 = new Photo("Photo 6", "Constantin", 5, 15);
        Photo photo7 = new Photo("Photo 7", "Constantin", 5, 15);
        Photo photo8 = new Photo("Photo 8", "Constantin", 5, 15);

        article1.addPhoto(photo1);
        article1.addPhoto(photo2);
        article1.addPhoto(photo3);

        article2.addPhoto(photo4);

        article3.addPhoto(photo5);

        ad1.addPhoto(photo4);
        ad1.addPhoto(photo5);
        ad1.addPhoto(photo6);

        ad2.addPhoto(photo6);
        ad2.addPhoto(photo7);

        ad3.addPhoto(photo8);

        Newspaper nouvelliste = new Newspaper("Nouvelliste", "20.02.2009",
                                                3, 3, 2);

        nouvelliste.addArticle(article1);
        nouvelliste.addArticle(article2);
        nouvelliste.addArticle(article3);
    }
}
```

```

        nouvelliste.addArticle(article4) ;

        nouvelliste.addAd(ad1);
        nouvelliste.addAd(ad2);
        nouvelliste.addAd(ad3) ;

        System.out.println(nouveliste);
    }
}

```

When you execute the Test class you should obtain this result on the console:

```

The article 'Titre 1' is full you can't add a new photo
The ad 'Gruyeres AOC' is full you can't add a new photo
The newspaper is full you can't add a new article
The newspaper is full you can't add a new ad
Newspaper :
Name : Nouvelliste
Date : 20.02.2009
Number of pages : 3
Articles :
Author : Zuber
Title : Titre 1
Text : BlaBla 1
Photo :
Name : Photo 1
Photograph : Constantin
Width : 5
Height : 15
Photo :
Name : Photo 2
Photograph : Constantin
Width : 10
Height : 13

Articles :
Author : Emery
Title : Titre 2
Text : BlaBla 2
Photo :
Name : Photo 4
Photograph : Constantin
Width : 9
Height : 15

Articles :
Author : Zufferey
Title : Titre 3
Text : BlaBla 3
Photo :
Name : Photo 5
Photograph : Constantin
Width : 5
Height : 15

Ads :
Ad :
Author : Ruedin
Title : Gruyeres AOC
Text : BlaBlaBla
Company : Migros

```

```
Photo :  
Name : Photo 4  
Photograph : Constantin  
Width : 9  
Height : 15  
Photo :  
Name : Photo 5  
Photograph : Constantin  
Width : 5  
Height : 15
```

```
Ad :  
Author : Zuber  
Title : NF  
Text : BlaBlaBla  
Company : Nouvelliste  
Photo :  
Name : Photo 6  
Photograph : Constantin  
Width : 5  
Height : 15  
Photo :  
Name : Photo 7  
Photograph : Constantin  
Width : 5  
Height : 15
```

Ad.java,
Article.java,
Newspaper.java,
Photo.java,
TestNewspaper.java

W2Q3 – 3 Occurence

OccurrenceString.java
OccurrenceMain.java